08 | ViT local operators: full LayerNorm and GELU derivatives

Local analytic examples supplement the patch-attention-residual chain; no trained-model measurements.

$$\frac{\partial y_i}{\partial u_j} = \frac{\gamma_i}{\sigma}\left[\delta_{ij} - \frac{1}{D} - \frac{\hat{u}_i \hat{u}_j}{D}\right]$$

$$GELU'(u) = \Phi(u) + u\phi(u)$$

Full LN: target y1

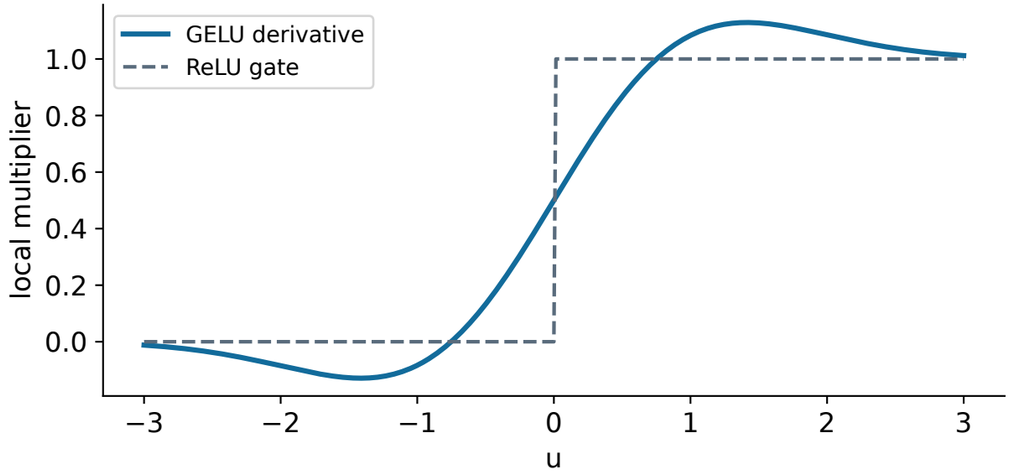
-0.400	-0.081
0.181	0.165

$\Sigma = -0.1344$

Frozen mean / variance

-0.813	0.000
0.000	-0.000

$\Sigma = -0.8131$



$u = [-1,\ 0.5,\ 2,\ -0.5], \quad D = 4, \quad \varepsilon = 0.2$ (maps in local u coordinates)

LN uses the biased variance (division by D). Mean and variance couple channels.

LN beta, projection bias, CLS and positional parameters affect downstream states; they have no direct image seed.

GELU uses its continuous endpoint derivative. Substituting a binary gate changes the model Jacobian.

A full ViT implementation retains these operators, both residual paths, head concat and output projection in eval mode.